

Day 3: Configuring RIP Theory: RIP concepts

Task 1 :- Set up a network topology with three routers in Cisco Packet Tracer, connect them using serial or Ethernet links, and assign IP addresses to all interfaces.

ANS :-

- 1.Place Routers: Drag 3 routers (R1 — R2 — R3) into Packet Tracer.
- 2.Cable Them: Connect R1 → R2 and R2 → R3 using Copper Cross-Over (or Serial) cables.
- 3.Set R1 IP:

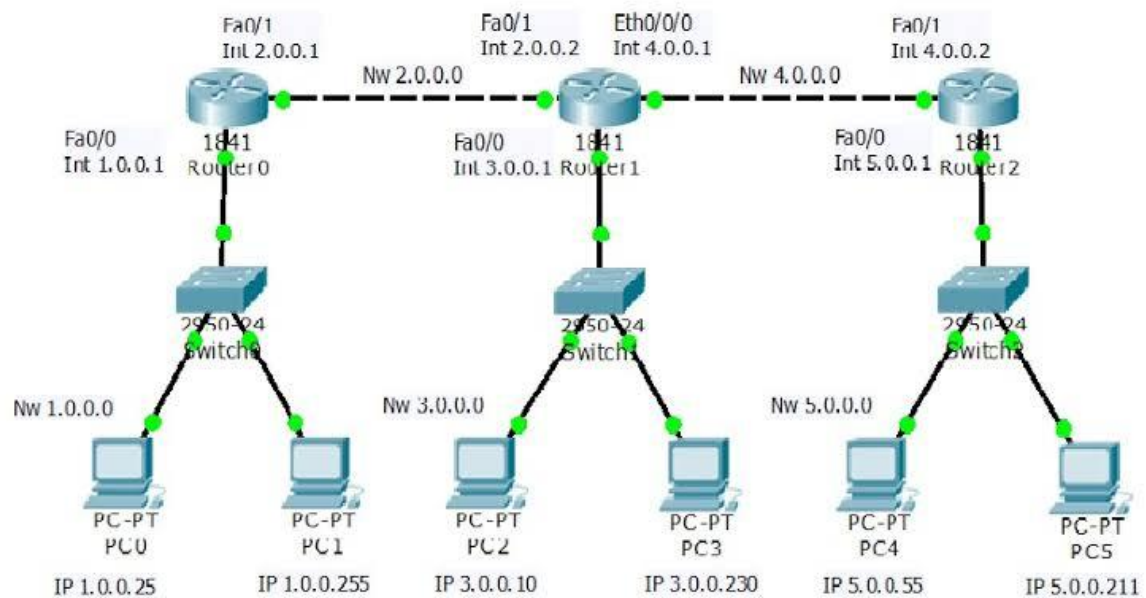
```
en
conf t
int g0/0/0
ip addr 10.1.1.1 255.255.255.0
no shut
```

- 4.Set R2 IPs:

```
en
conf t
int g0/0/0
ip addr 10.1.1.2 255.255.255.0
no shut
exit
int g0/0/1
ip addr 10.2.2.1 255.255.255.0
no shut
```

5.Set R3 IP:

```
en
conf t
int g0/0/0
ip addr 10.2.2.2 255.255.255.0
no shut
```



All link lights turn green once both ends are turned on

Task 2 :- Enable RIP version 2 on all three routers in your topology, advertise all directly connected networks, and verify that each router learns routes to all subnets using the 'show ip route' command.

ANS :-

1. Enable RIPv2 on Each Router

Open each router's CLI and run these commands:

On R1:

```
en
conf t
router rip
version 2
no auto-summary
network 10.1.1.0
```

On R3:

```
en
conf t
router rip
version 2
no auto-summary
network 10.2.2.0
```

2. Verify Learned Routes

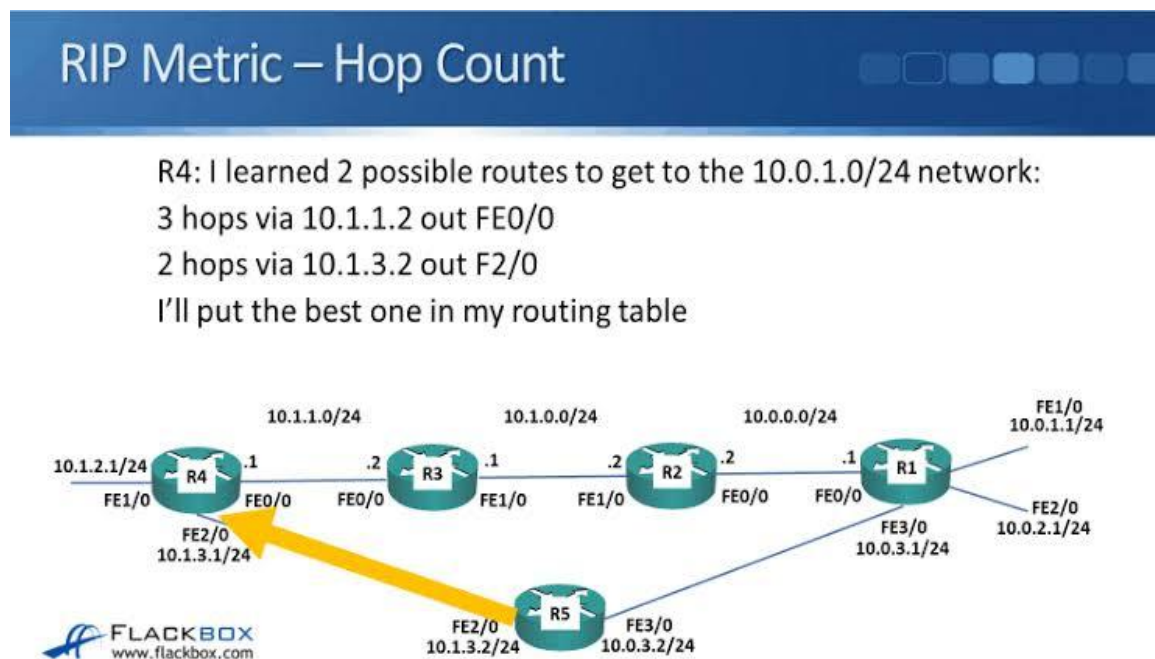
Run this command on any router:

```
show ip route
```

Success check: Look for lines starting with **R**. If you see **R** next to the remote IP networks, RIPv2 is working!

Task 3 :- Change the hop count metric for one of the networks by adding an extra router between two existing routers, then observe and record how the hop count value changes in the routing tables.

ANS :-



1. Add a 4th router (R4) between R1 and R2.
2. Connect and set IPs for the new links (R1 → R4 → R2).
3. Enable RIPv2 on R4 (router rip, version 2, network ...).

What to Observe & Record

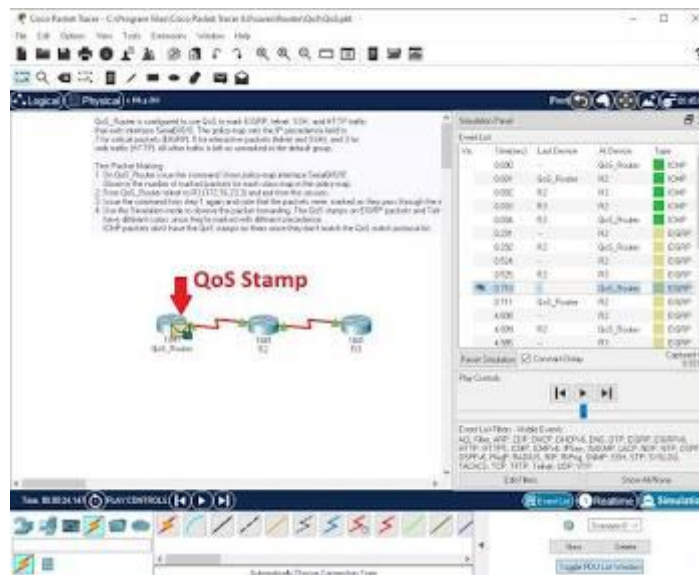
1. Run show ip route on R1 or R3.
2. Hop count increases by +1:
 - Before: [120/1] (1 hop away)
 - After: [120/2] (2 hops away through the extra router)

(RIP uses **hop count** as its metric—each additional router passed increases the value by 1).

Task 4 :- Use the simulation mode in Cisco Packet Tracer to capture and analyze RIP update packets as they are exchanged between routers.

ANS :-

Steps to Capture & Analyze RIP Packets :-



Switch Mode : Click Simulation in the bottom-right corner (or press Shift + S).

Filter RIP Only : Click Edit Filters → Uncheck everything → Under the IPv4 tab, check RIP.

Capture Packets : Click the Play / Forward button (>) to let Packet Tracer step through network events.

Inspect Packet : Click any colored envelope (RIP packet) on the workspace or in the Event List.

View Details :

- **Inbound/Outbound PDU Details :** Check the multicast destination IP (224.0.0.9 for RIPv2).
- **RIP Header :** View advertised routes, subnets, and metric/hop counts inside the payload.